My Portfolio!

George Kopf V

Electrical & Computer Engineering | Princeton University
Portfolio: gkopf5.github.io | 609-608-2119
georgekopf5@princeton.edu | in/george-kopf-v-a0b895270/

Education

Princeton University, Princeton, NJ
B.S.E. Electrical & Computer Engineering, Minor in Robotics – 3.6 GPA
Aug 2023 – May 2027 (Expected)

Professional Experience

Undergraduate Researcher / Research Assistant — Princeton University BAM Lab
Feb 2026 – Present

- Designed and fabricated a robotic model organism of *Exocoetus* flying fish with integrated electrical control and measurement systems, enabling 2.4 GHz wireless telemetry at 50 Hz and untethered swimming speeds of 11 cm/s at a 2 Hz gait frequency; current work involves pectoral fin deployment and feedback control.
- Executed comprehensive hardware integration, including custom PCB design, electromechanical assembly, wireless communication protocols, and waterproofing.

Research Assistant — Kent State University REU
Jun 2025 – Aug 2025

- Characterized and debugged an adjoint-based gradient descent optimization algorithm used to simulate and optimize fish swimming gaits.
- Developed and validated custom LabVIEW data acquisition and control software for an experimental fluid dynamics apparatus measuring water flow through deformable tubing.

Research Assistant — Princeton University iGEM Team
Jun 2024 – Aug 2024

- Engineered probiotic bacterial therapies for ulcerative colitis and designed and fabricated gel-based microfluidic test chambers for quantitative biological response measurement; contributed to the 2024 iGEM Gold Medal project.

Course Projects & Extracurriculars

TigerCub: PocketQube Satellite

- Designed, built, and tested a custom burn-wire deployment PCB for a deployable retroreflector on a 1P PocketQube satellite, switching 1 A at 5 V for 5 seconds (MAE 348: Spaceflight Projects).
- Performed hardware integration, verification, and electrical testing on an open-source DynOSSAT-EDU EPS module, verifying solar panel power distribution and battery charging while maintaining regulated 3.3 V/5 V outputs under a 1 A load.

Self-Driving Car

- Developed a custom dual-layer iLQR safety filter for real-time lane and obstacle detection on an RC testbed vehicle; integrated Python, machine learning models, and ROS 2 (ECE 346: Intelligent Robotics).

Search and Rescue Robot

- Served as electrical lead on an 11-person team, designing and assembling the complete power and signal wiring harness for an autonomous robot, distributing 12 V and 9.6 V power across 12 components (MAE 322: Mechanical Design).

Princeton Racing Electric (FSAE & F+HE)
Sep 2024 – Present

- Directed the water-cooling subteam, contributed to 600 V high-voltage safety architecture, and manufactured structural carbon-fiber composite components using Siemens NX for 3D modeling and thermal validation.

Technical Skills

- **Hardware/CAD:** SolidWorks, Siemens NX, Fusion, Creo, Altium, KiCad, EasyEDA
- **Software:** Java, Python, C, C++, Arduino, LabVIEW, PSoC Creator, MATLAB, Verilog
- **Instrumentation:** Oscilloscopes, DMMs, AFGs, spectrum analyzers, PLCs, CubeSat environmental testing
- **Machining:** Trained on heavy machine-shop equipment, including lathes, mills, and CNC machining

Relevant Coursework

- **Robotics:** Robotics/Autonomous Systems, Intelligent Robotics, Power Electronics, Mechanical Design
- **Miscellaneous:** Advanced Wireless Systems, Fluid Dynamics, Spaceflight Projects
- **Upcoming:** Differential Equations, Space System Design